

Traffic Pattern Analysis

How we simulate the three sub-problems

Validation design — no results in this deck

What has to be predicted

The deliverable, and what it is really asking

Fill `submission_template.csv`

170,956 rows, each needing a point forecast, a 90% interval, and a reliability score — plus 5 scenario answers.

Scoring

Forecast performance	35%
Scenario analysis	20%
Uncertainty & reliability	15%
Robustness	15%
Reproducibility	10%
Technical quality	5%

The trap

Only **35%** rewards point accuracy. Half the score rewards behaviour on the *hard slices* and honest uncertainty.

So the first job is not a model. It is working out *which* rows are hard, *why*, and building a way to measure that honestly.

What we are given, and what is withheld

File	Period	Content
traffic_train.parquet	2024	labelled — 75 stations, 902,902 rows
traffic_validation.parquet	Jan–Jun 2025	labelled — <i>same</i> 75 stations, 556,800
network.csv + _edges.csv	—	127 stations, 186 typed edges
submission_template.csv	2024-01 – 2025-12	withheld — 118 stations, 170,956

Two facts that shape everything that follows:

- validation is *not* a validation set. It is the same 75 stations, six months later. For the final model it is training data.
- Of the **118** stations we must predict, only **75** appear in any labelled file. **43 have no history anywhere** — 29% of the rows to predict.

Consequence

No metric can ever be computed on the real target rows — their labels are withheld. Everything we report comes from a *simulation* built from the 75 stations we have.

The target file is three different statistical problems

Slice	Rows	Information available at prediction time	Nature
warm-future	120,571 (70.5%)	own history to 2025-06-30, calendar, road attributes	extrapolate in time
cold-interp	23,810 (13.9%)	no own labels ever, but ~68 stations observed at the target hour	interpolate in space
cold-future	26,575 (15.5%)	no own labels, and nothing observed anywhere at that hour	both at once

Verified directly from the files:

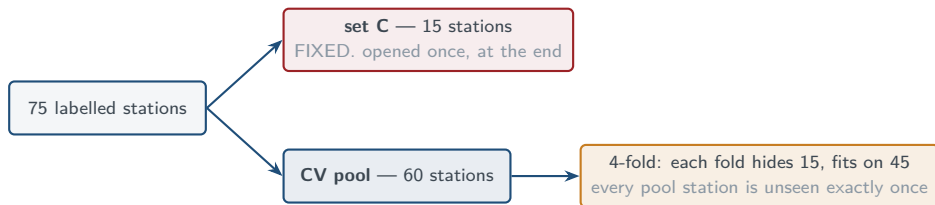
- **no target row overlaps any label;**
- for all 147,146 future rows, the number of stations readable at the target timestamp is **exactly zero**;
- for the 23,810 backfill rows the median is **68 stations readable at that same hour**, and **99.7% of them fall in Jan–Jun 2025**.

How we simulate all three

The idea: ignore the target file completely

The only honest move available

The 43 truly-unseen stations have no labels. So we take the **75 that do**, hide some from ourselves, and measure how well we recover what we threw away.



Why nested, and why k-fold

A single 15-station holdout carries ± 3.5 points of $\text{GEH} < 5$ — enough for two models to swap places by luck. Rotating folds puts every pool station in the unseen role once, roughly halving that noise. Set C stays out: if every station were used for selection, nothing would be left for an unbiased final number.

The three experiments, on one grid

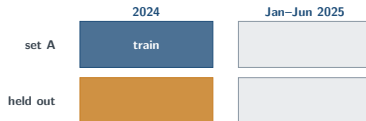
1. warm-future



Predict set A's **2025 H1** — a station we know, a period we do not.

= real **warm-future**: 70.5% of the rows to deliver.

2. cold-interp



Reconstruct the held-out stations' **2024** — **inside** the window, so neighbours are recording.

= real **cold-interp**: never-seen station, neighbours *were* recording.

3. cold-future



Forecast the held-out stations' **2025 H1** — **after** the window, nothing recording anywhere.

= real **cold-future**: unknown station, unknown period.



Inside the labelled window vs **after** it — the same relationship the real backfill and future targets have to the real labelled window.

Why this mirrors the real task

What matches

Real task	Simulation
history → future	2024 → 2025 H1
backfill <i>inside</i> window	same
future <i>after</i> window	same
6-month horizon	same
75 seen / 43 unseen	45 / 15
nbrs observed in backfill	same
nothing observed in future	same

What does not match — both make us look worse

Calendar months. The real target is Jul–Dec. We score Jan–Jun, which contains the two hardest months on this network.

Drift is unestimable. The year-over-year level shift is learned by matching months across years. Training on 2024 alone, there is nothing to match — so the simulation forecasts 2025 with no drift correction, while the deployed model measures it from Jan–Jun 2025 first.

Reading the numbers

The protocol is deliberately **conservative**. Treat its output as a lower bound on real-task performance, not an estimate of it — and never quote a number from an easier split alongside it.

Confirmed with DeIDOT — and one constraint we were breaking

Confirmed by Sai Ram Nomula

Batch forecast. Jul–Dec 2025 uses information through **2025-06-30**. No new observations are revealed during the horizon.

Planning, not operations. The objective is *expected hourly volume for planning*, not real-time incident prediction — which is why a model-calibration standard like GEH is the right metric.

Model-generated history is allowed for stations with no observations, built from neighbours and network relationships — but *“treated as an estimate rather than observed truth”*.

Limited history is deliberate: *“part of the analysis is determining whether an approach can generalize beyond locations with long, complete histories”*.

The prediction-time rule

“Use only information that would have been available at that prediction time.” Fitting on 2024 and predicting Jan–Jun 2025 satisfies this. It also means **cold-start generalisation is explicitly part of what is being judged** — so the cold slices deserve more effort than their 29% row share suggests.

Can the reconstruction feed the forecast?

The idea

Experiment 2 reconstructs a never-seen station's 2024; experiment 3 must forecast that same station's 2025. So hand the reconstruction to the forecaster as history. DelDOT permit this explicitly; it is the KITS / MoGERNN pattern.



Two arms, and let the numbers decide

Reconstruction is *smooth* — no weather, no incidents, no sensor noise — so a model trained on it sees less residual variance than reality and its intervals come out too narrow, on exactly the stations it knows least. Report **direct** and **chained** side by side; fit intervals only on real held-out data.

How we score it

GEH, not MAE

The traffic-engineering standard

$$\text{GEH} = \sqrt{\frac{2(M - C)^2}{M + C}}$$

M modelled, C counted. Self-normalising: forgiving of small absolute errors on quiet roads, strict on busy ones.

GEH < 5	acceptable on a link
5–10	needs investigation
> 10	poor fit

Industry target: $\geq 85\%$ of links below 5

Why MAE alone is not enough

“MAE 142” says nothing about whether the model is *useful*. GEH says whether a DOT would accept the calibration.

Reported alongside it:

- % of rows within $\pm 20\%$ / $\pm 30\%$
- R^2 — variance explained
- empirical 90% coverage and Winkler score, for the interval
- % of the gap between a naive baseline and the achievable ceiling

Every one of these is computed separately for warm-future, cold-interp and cold-future — a single blended number would hide exactly the slices that carry half the score.

Summary — what gets measured, and against what

Experiment	Fit on	Predict	Scored against	Stands in for
1. warm-future	seen, 2024	seen, 2025 H1	real labels	120,571 rows (70.5%)
2. cold-interp	seen, 2024	held out, 2024	real labels	23,810 rows (13.9%)
3. cold-future	seen, 2024 $\pm$ recon	held out, 2025 H1	real labels	26,575 rows (15.5%)

The rules this respects

Only information available at or before the predicted timestamp — including inside the backfill window.
No random k -fold: it would leak time, station behaviour and outage context. Cold experiments hide *whole stations*, never random cells.

Still open

The **5 scenario prompts** are not in the supplied data. They carry **20%** of the score, and no defensible answer exists without them.

Questions